

House Price Prediction

Project Summary

Objective

I want to build an ML model that will predict the house prices based on its features such as area, number of bedrooms/bathrooms, parking, furnishing, air condition, etc. I also want to find out which feature affects the price the most.

Dataset and Data Preparation

I am using the Housing Prices dataset for my project. First, I checked if there are any missing value and duplicates in my data set. After that I converted the column having values into numbers by one-hot encoding.

Model Building

We split our dataset into 80% Training data and 20% Testing data and built two Machine Learning Models: Linear Regression and Random Forest Regressor. We then compared their performance by R^2 score and Prediction Error.

Results & Insights

From the heatmap, we found that area is a factor influencing house prices. Houses with larger areas generally have higher prices. Amenities like AC and parking also showed an impact on property prices. Our Linear Regression model achieved an R^2 score of 0.65.

The predictions were off by around ₹9.7 lakhs on average, but the model was still able to estimate house prices reasonably well. One interesting finding was that although Random Forest Regressor usually performs better, Linear Regression actually did better on this dataset, showing that there is a direct relationship between our chosen features and house prices.

Overall, this project helped me understand the entire machine learning workflow, and based on the results, real estate businesses can focus on properties and offer amenities like AC and parking facilities to increase their property value and draw in more buyers.